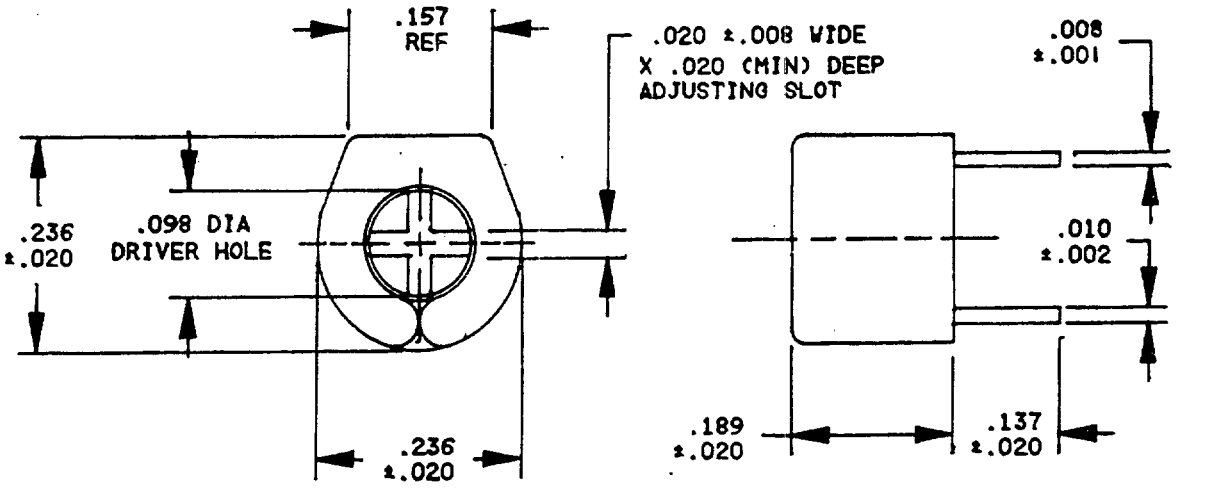
MILITARY SPECIFICATION SHEET

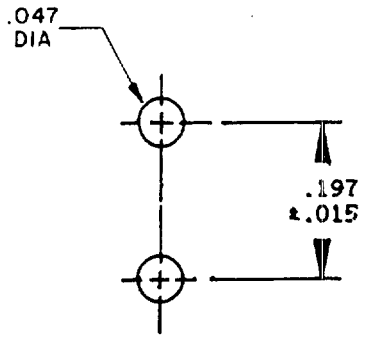
CAPACITORS, VARIABLE, CERAMIC DIELECTRIC,

STYLE CV42

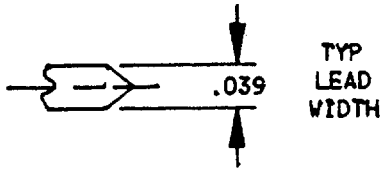
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification sheet and the latest issue of MIL-C-81.





RECOMMENDED
MOUNTING HOLES



INCHES	MM	INCHES	MM
.001	0.03	.047	1.20
.002	0.05	.098	2.50
.006	0.15	.137	3.50
.008	0.20	.157	4.00
.010	0.25	.189	4.80
.015	0.40	.197	5.00
.020	0.50	.236	6.00
.039	1.00		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.006 (0.15 mm).

FIGURE 1. Style CV42 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

DC rated voltage: 100 volts dc.

Operating temperature range: -55°C to +85°C.

Characteristic: See table I.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. A dc potential of 2.2 times rated voltage applied between terminals for 3 ±2 seconds.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 ft.).

Test potential: 100 percent of dc rated voltage.

Insulation resistance: Method 302 of MIL-STD-202, condition A, 100 volts dc applied - 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: At 1 MHz ±100 kHz, at maximum and minimum capacitance - shall not be more than 0.2 percent for rated maximum capacitance 11 pF, 20 pF, and 30 pF, 0.3 percent for 7 pF.

TABLE I. Style CV42.

Type designation	Capacitance		DC rated voltage	Characteristic	
	Min	Max		Symbol	Temperature coefficient
	pF	pF	volts		ppm/°C
CV42X070	2.0	7.0	100	X	0 ±200
CV42Y110	3.0	11.0	"	Y	-450 ±300
CV42Z200	4.2	20.0	"	Z	-750 ±300
CV42Z300	5.2	30.0	"	Z	-750 ±300

Temperature coefficient: Within the limits specified for the applicable characteristic.

Capacitance drift: Within 0.50 picofarad (pF) of initial step 1 measurement.

Terminal strength:

Pull test: Capacitor held by body and 4-pound load applied to each terminal for at least 10 seconds.

Torque: Not less than .3 nor more than 2 ounce-inches.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 G).

Capacitance change: Shall not exceed ±2 percent or 0.5 pF, whichever is greater.

Dissipation factor: Shall be not more than 0.2 percent for rated maximum capacitance 11 pF, 20 pF, and 30 pF, 0.3 percent for 7 pF.

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Dielectric withstanding voltage: A dc potential of 2.2 times rated voltage applied between terminals for 3 ±2 seconds.

Insulation resistance: 10,000 megohms, minimum.

Fatigue:

ΔC: Shall not exceed 12 percent or 0.75 pF, whichever is greater.

Torque: Initial requirement.

Life:

Qualificaton test: 1,000 hours at 85°C, 150 percent of rated volts dc with a peak alternating voltage of 50 percent of rated volts dc (100 hertz or less) superimposed.

Insulation resistance: Initial requirement.

Capacitance change: Shall not exceed ±8 percent or 0.75 pF, whichever is greater.

Group C life: Conditions and requirements are the same as that for qualification.

Moisture resistance: Method 106 of MIL-STD-202.

Insulation resistance: 10,000 megohms minimum.

Capacitance change: Shall not exceed ±5 percent of nominal value or 0.5 pF, whichever is greter.

Dissipation factor: Shall be not more than 0.5 percent.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Air Force - 11, 80, 99
DLA - ES

User activities:
Army - MI
Air Force - 19
Navy - MC

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5910-1482)

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